Grounding-Preserving Graph Routing for Reliable Multimodal Web Understanding

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Abstract

Web pages contain textual content, visual layout and structural relations. Both web reading and operation depend on taking these signals as joint evidence rather than as isolated cues. Integrating these signals therefore plays an important role in reliable multimodal web understanding and grounding. Existing methods typically supply structural evidence through visual grounding on screenshots, serialized document trees, or interface graphs as attention bias. However, challenges are still posed by untraceable evidence sources across layout and tree views, shifted point coordinates under structural writes, and implicit channel reliability across disagreeing representations. To address these challenges, we propose GPGR, a grounding preserving graph routing framework for reliable multimodal web understanding. Specifically, GPGR models each page as a dual view heterogeneous graph with separate spatial and tree channels and passes typed messages over budgeted anchors at linear cost. In addition, a grounding preserving orthogonal injection projects visual messages into the orthogonal complement of the grounding subspace and yields an exact invariance guarantee. Furthermore, a reliability aware routing mechanism flips one channel per relation during training and learns a gate whose optimum equals the resulting reliability. Extensive experiments are carried out on three public benchmarks, indicating that GPGR improves multimodal web understanding over representative baselines while preserving the grounding ability of the frozen backbone.

Keywords

Multimodal Large Language Models, Structural information, Web Page Understanding, Visual Grounding

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1 Introduction

Web pages are the most common multimodal interface on the web, and they simultaneously convey textual content, two-dimensional visual layout, and structural relations such as containment, sibling order, and the anchoring between a text entity and the image it describes. Relational judgment underlies both web reading and web operation, since an answer or an action often depends on which element contains a given widget or which sibling comes first in the document tree. This raises a central question: *How can multimodal large language models (MLLMs) integrate structural relations with textual content and visual layout, to achieve reliable web understanding and grounding?*

To address this question, three routes have been explored to supply structural information to a web understanding model. The pure visual route trains a grounding head on the screenshot alone. CogAgent [9], SeeClick [3], and OS-Atlas [29] achieve strong point grounding but supply no explicit structure. GUI-Actor [28] and UI-TARS [19] extend this route to agents, yet they leave tree order and containment without an information source. The screenshot with serialized structure route feeds a linearized page tree into the prompt. SeeAct [34], WebAgent [8], and HtmlRAG [22] feed the document object model (DOM) or a distilled tree into text, yet spatial relations and sibling order are lost in the serialization. The interface graph modeling route builds a graph from the page. WebFormer [26], Graphormer [31], and ShowUI [13] use it for token pruning and attention bias rather than for injecting typed relation messages as a signal that can be routed by type.

Despite these advances, existing methods still face three technical challenges when integrating structural relations into MLLMs. First, the layout and the document tree form two views of the same page under separate identities. Combining them into a single representation, however, obscures the origin of each piece of evidence. A representation that keeps the two views separate and traceable would address this, yet such a representation requires propagation over the token graph, and a full propagation over every token pair is quadratic in the page length, which restricts the approach to short pages. Second, the backbone already achieves accurate point grounding on plain screenshots. Adding structural messages to the frozen visual states can shift the coordinates the model predicts from these states, so the new evidence arrives at the cost of a capability that already works. The point coordinates and the relational evidence share the same directions in the frozen states. A uniform scale attenuates all of them together, sacrificing the useful relations with the harmful ones. Third, relations of different types are carried by different representations, and these representations may disagree on the same page. The trustworthy source for each relation stays implicit in the training data, which surfaces only the final answer. Training on the answer alone mixes task performance with the trustworthiness of the representations, and the model can reach the right answer through the wrong source.

Motivated by these challenges, we present GPGR, a training-efficient graph routing framework for frozen multimodal backbones. Specifically, GPGR models the page as a dual-view heterogeneous graph with a spatial region view and a structural tree view. The two views are aligned by entropy-regularized optimal transport without learnable parameters, keeping the source of each piece of evidence traceable, and typed message passing over a budgeted set of region

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anchors produces three source-preserving channel messages at linear cost [4]. The grounding subspace is estimated by singular value decomposition on the visual states of plain screenshots, and visual messages are projected into the orthogonal complement of this subspace while text messages are left unchanged, so the point coordinates stay intact. We prove an exact grounding-invariance result together with a minimal perturbation guarantee and a leakage bound across layers. A counterfactual intervention instrument flips only one channel of evidence for each relation during training, and the channel reliability is defined as the probability of following the flip. A gate routes each relation type by reliability, and we prove that the optimal gate equals the reliability with a convergence rate. The instrument is removed at inference, so the gate adds no serving cost. A concurrent diagnostic study attributes the grounding collapse to a channel belief mismatch, and our instrument generalizes its binary attribution into a graph-level intervention with a trained gate [33].

The contributions of this paper are summarized as follows.

- We formalize the web page as a dual-view heterogeneous graph with a spatial region view and a structural tree view, and we introduce a matched representation family that keeps the two views in separate message channels.
- We propose grounding-preserving orthogonal injection that projects structure messages into the complement of the grounding subspace, and we prove an exact invariance guarantee with a bound on the residual leakage across layers.
- We propose reliability-aware counterfactual routing trained from paired graph interventions, and we prove that the optimal gate equals the channel reliability with a convergence rate.
- We evaluate GPGR under a matched fairness pool on three benchmarks with two frozen backbones, report reading, element, action, and text-versus-icon grounding separately, and provide a mechanism analysis explaining why each component is necessary.

2 Related Work

2.1 Multimodal Web Understanding

Multimodal web understanding has emerged as a central research direction with the rapid advancement of multimodal large language models, and existing methods can be broadly divided into two categories according to the modality through which structural evidence is supplied. The pure visual route encodes only the rendered screenshot and relies on the backbone to infer structure from pixel layout. Hong et al. proposed CogAgent [9], a visual language model that integrates high-resolution cross-attention with screenshot inputs for GUI agents. Cheng et al. introduced SeeClick [3], which harnesses GUI grounding to advance visual GUI agents through coordinate supervision. Wu et al. proposed OS-Atlas [29], a foundation action model that learns generalist GUI agents across platforms, and further proposed GUI-Actor [28], a coordinate-free visual grounding method for GUI agents. Qin et al. introduced UI-TARS [19], which pioneers automated GUI interaction with native agents. These models rely on large-scale pretraining and coordinate supervision to localize elements, and they achieve strong point grounding on GUI benchmarks, yet they lack an explicit information source for tree order and containment, so relational questions whose evidence lies in the document tree cannot be answered from the rendering alone.

The structure-as-text route serializes the page tree into a prompt through a DOM string, an accessibility tree, or a graph linearization. Zheng et al. proposed SeeAct [34], which shows that GPT-4V(ision) serves as a generalist web agent when grounded. Gur et al. introduced WebAgent [8], a real-world web agent with planning, long context understanding, and program synthesis. Tan et al. proposed HtmlRAG [22], arguing that HTML is better than plain text for modeling retrieved knowledge in RAG systems. Liu et al. introduced MultiSUI [14] to harness webpage UIs for text-rich visual understanding, and Fatemi et al. proposed Talk-like-a-Graph [6] to encode graphs for large language models. Lamott et al. proposed LAPDoc [12], a layout-aware prompting approach for documents, and Zhou et al. introduced WebArena [35], a realistic web environment for building autonomous agents. These methods trade spatial relations for tree relations that the text channel can recover. Benchmarks such as WebSRC [2], VisualWebBench [15], and Mind2Web [5] quantify the gap between the two routes on reading, element, and action tasks, while a concurrent diagnostic study [33] attributes the residual errors to a channel belief mismatch between pixels and structure.

2.2 Graph Modeling of Web Pages and GUIs

Graph modeling of web pages and GUIs has been investigated along two complementary lines. The first line concerns graph neural networks for structured data in general. Veličković et al. proposed Graph Attention Networks [24], which introduce typed attention over neighbor features and have become a standard operator for propagating evidence on graphs with learned weights. Schlichtkrull et al. introduced Relational Graph Convolutional Networks [21], extending the convolution to multiple edge types so that heterogeneous relations can be modeled within a single message-passing framework. Ying et al. proposed Graphormer [31], which adapts the Transformer to graph-structured inputs through structural encodings and reaches strong performance on a range of graph benchmarks. These methods establish the foundation for propagating typed messages over a region graph with a bounded receptive field, and they have been adopted in recommendation, document, and web tasks where relational structure is the primary signal.

The second line concerns web-specific graph construction. Wang et al. proposed WebFormer [26], a web-page Transformer for structure information extraction that builds a page-level encoder over DOM nodes. Yuan and Färber introduced GraSAME [32], which injects token-level structural information into pretrained language models through a graph-guided self-attention mechanism. Kil et al. proposed Dual-VCR [11], which contextualizes web navigation through neighbor relations over a region graph. Fatemi et al. further proposed Talk-like-a-Graph [6], and Perozzi et al. introduced GraphToken [18], both of which supply graph structure to large language models through serialization. These encodings keep the graph explicit but lose the spatial channel through total ordering, and they have been evaluated on a range of structure-rich benchmarks where the graph structure is the primary signal.

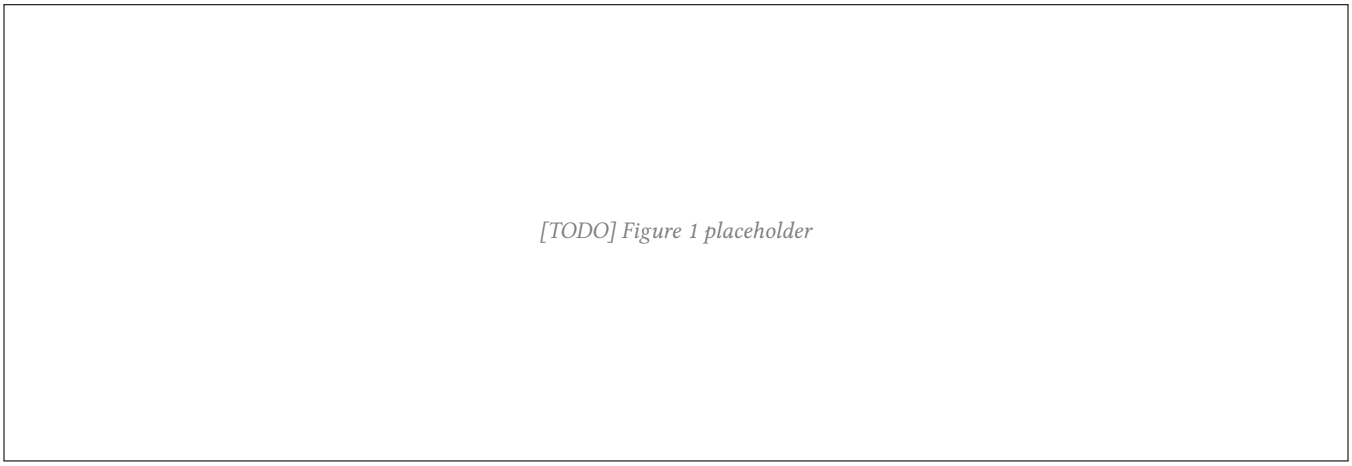


Figure 1: Framework overview of the proposed GPGR.

2.3 Injecting Structural Information

Injecting structural information into pretrained models has been explored through several lines of work, which differ in where the structure enters the model. Wang et al. proposed DocLLM [25], a layout-aware generative language model for multimodal document understanding that injects bounding-box coordinates through disentangled attention matrices. Lamott et al. introduced LAPDoc [12], which verbalizes region boxes as layout-aware prompts for documents. Kil et al. proposed Dual-View [11], which contextualizes neighbor regions through cross-attention over a region graph. Yang et al. introduced Set-of-Mark prompting [30], which overlays numeric labels on the screenshot so that the backbone can refer to elements by their marks. Liu et al. further proposed MultiSUI [14] to harness webpage UIs for text-rich visual understanding through serialized structure, and these layout-aware methods collectively show that explicit spatial cues improve reading and grounding on structure-rich benchmarks.

Graph-guided approaches further integrate structural evidence into the model computation. Yuan and Färber proposed GraSAME [32], which injects token-level structural information through a graph-guided self-attention mechanism. Wang et al. introduced WebFormer [26], which builds a page-level Transformer that consumes DOM nodes directly. Hu et al. proposed LoRA [10], which enables parameter-efficient adaptation of large models and has become a standard recipe for adapting frozen backbones with structure modules. These methods span prompt-only approaches, module-level adaptations under a shared LoRA budget, and graph-guided attention variants, and they collectively show that explicit structure improves reading and grounding on structure-rich benchmarks.

3 Method

3.1 Framework Overview

We formulate relational webpage understanding as structure injection into a pretrained multimodal large language model M, subject to preserving its grounding capability. Given a webpage screenshot and a query q, the vision Transformer of M encodes the screenshot into visual tokens, which are processed together with the query by the language model to produce hidden states $H \in \mathbb{R}^{n\times d}$. For the analysis, we consider a linear grounding head $\ell_g(H) = HW_g$, where $W_g \in \mathbb{R}^{d\times 2}$ maps each relevant hidden state to a point in the screenshot. Our objective is to incorporate page structural information to improve web understanding while preserving the expected output of this head. To this end, we represent the webpage through two complementary graph views, denoted jointly by $G = (G^{\text{spa}}, G^{\text{str}})$. The spatial graph G^{spa} is derived from the rendered layout, while the structural graph G^{str} follows the document tree.

Figure 1 illustrates how the graph information is incorporated into the frozen backbone. Starting from the spatial and structural views constructed in Section 3.2, the plugin computes a separate message channel from each view and encodes a serialization of the nodes and edges of G to form a third channel following Fatemi et al. [6]. These channels provide complementary spatial, structural, and textual evidence. To control how this evidence modifies the visual states, Section 3.3 projects visual messages onto the orthogonal complement of a protected subspace estimated from frozen visual states. For the linear grounding head, exact local preservation requires this subspace to contain the column space of W_g, while approximate coverage leads to residual leakage. The projected messages are then combined by the router in Section 3.4, which uses controlled interventions during training to learn which channel to favor for each relation type. The intervention analysis assumes independent pairs stratified by relation type. The resulting residual $\Delta_\theta(G)$ is added to H, yielding the prediction $\hat{y} = f_\theta(M, G, q)$, where θ denotes the trainable plugin parameters and the backbone remains frozen. We suppress the dependence of $\Delta_\theta(G)$ on the hidden states and query for notational simplicity. Let $\mathcal{D}$ denote the distribution over page graphs, queries, and answers, with the screenshot associated with each graph included in the model input. We define the expected answer loss as $\mathcal{L}_{\text{task}}(\theta) = \mathbb{E}_{(G,q,y)\sim\mathcal{D}}[-\log p_\theta(y \mid M, G, q)]$. The training objective in Section 3.5 combines answer supervision with auxiliary losses to support the constrained goal

$$\min_\theta \ \mathcal{L}_{\text{task}}(\theta) \ \text{ s.t. } \ \mathbb{E}_{G\sim\mathcal{D}}\big[\ell_g\big(H + \Delta_\theta(G)\big)\big] = \mathbb{E}_{G\sim\mathcal{D}}\big[\ell_g(H)\big]. \quad (1)$$

This constraint expresses the goal of improving answer prediction through structural evidence while preserving the grounding output in expectation. It does not require identical predicted points on every individual page.

3.2 Graph Construction and Soft Alignment

We construct two complementary graph views from a single rendering of each page. A spatial node $v \in V^{\text{spa}}$ represents a visible element with text z_v and normalized rendered box b_v. A structural node $u \in V^{\text{str}}$ represents a document element with text z_u, normalized box b_u, and attributes a_u. The node sets remain disjoint so that each view retains its identity. Spatial edges are determined by box geometry and use the relation set $\mathcal{R}^{\text{spa}} = \{\text{left}, \text{right}, \text{above}, \text{below}, \text{contain}, \text{overlap}\}$. Directional relations follow box centers, while containment and overlap follow box boundaries and intersections. Structural edges follow document nesting and use $\mathcal{R}^{\text{str}} = \{\text{parent}, \text{child}, \text{sibling}\}$, with sibling edges retaining the order of children under a common parent. These parameter free rules yield the spatial graph $G^{\text{spa}} = (V^{\text{spa}}, E^{\text{spa}})$ and the structural graph $G^{\text{str}} = (V^{\text{str}}, E^{\text{str}})$.

Although the two views describe the same page, their nodes need not share an identity. We therefore establish a soft alignment through a transport coupling between the n^{spa} spatial nodes and n^{str} structural nodes. Denote a and b as strictly positive probability vectors over the two node sets, and let $\Pi(a, b)$ contain all nonnegative matrices with row sums a and column sums b. The cost of matching v to u is $\Psi_{vu} = 1 - \langle \phi(z_v), \phi(z_u) \rangle + \|b_v - b_u\|_2^2$, where ϕ is a fixed text encoder whose outputs have unit norm. Thus a smaller cost indicates greater semantic and geometric agreement. The optimal coupling is

$$\hat{Q} = \arg \min_{Q \in \Pi(a,b)} \left[\langle Q, \Psi \rangle_F + \tau_Q \, \mathrm{KL}\big(Q \,\big\|\, a \otimes b\big) \right], \tag{2}$$

where $\tau_Q > 0$ controls the strength of regularization toward the independent coupling $a \otimes b$, defined by $(a \otimes b)_{vu} = a_v b_u$. The entry $\hat{Q}_{vu}$ defines the alignment mass between v and u. Under finite costs and positive marginals, the objective has a unique minimizer obtained by Sinkhorn scaling [4]. Appendix A.1 derives its equivalence to entropy regularization and the associated updates. The coupling is computed once per page before training and introduces no trainable parameters, aligning the views while keeping their node representations separate.

To bound the cost of subsequent message passing, we coarsen the spatial view into K region anchors with bounding boxes $\mathcal{B} = \{B_m\}_{m=1}^{K}$, where m indexes the anchors, $K \le K^{\max}$, and $K^{\max}$ is the anchor budget. Each box encloses the spatial regions merged into its anchor. For a spatial node v, define its overlap with anchor m as $\alpha_{vm} = |b_v \cap B_m| / |b_v|$, where $|\cdot|$ denotes area. The normalized assignment is

$$\Phi_{vm} = \frac{\alpha_{vm}}{\sum_{m'=1}^{K} \alpha_{vm'}}, \tag{3}$$

where $\Phi \in \mathbb{R}^{n^{\text{spa}} \times K}$ assigns spatial nodes to region anchors. If a node overlaps no anchor box, its assignment is set to one for the anchor with the nearest box center and zero for all others. Every row of Φ therefore sums to one, while a node overlapping multiple anchor boxes may distribute its assignment across them. Using this assignment, we transfer the alignment from spatial nodes to region anchors through

$$\tilde{Q} = \Phi^{\top} \hat{Q} \in \mathbb{R}^{K \times n^{\text{str}}}, \tag{4}$$

where $\tilde{Q}_{mu}$ is the transport mass connecting anchor m to structural node u. Spatial relations are also aggregated into relation matrices M_r between anchors, while structural relations remain at node resolution. Appendix A.2 gives the definition of M_r and derives the properties of both aggregations. Together, Φ, $\tilde{Q}$, and M_r support the attention operations in Section 3.3, with the anchors serving as compact carriers of spatial and aligned structural information.

3.3 Message Passing and Orthogonal Injection

This section turns the spatial and structural graphs into channel messages and writes them into the ViT hidden states H. The values that travel along edges come from node attributes. A visual hidden state is used only as a query when the message is written back, so the residual is structural evidence rather than a remix of H. Each channel is then projected onto the orthogonal complement of the grounding subspace. The channels stay separate here; the gate in Section 3.4 mixes them only after this projection, and any mixture of vectors in that complement remains inside it.

Spatial attributes are pooled onto anchors by the assignment of Section 3.2, $s_m = \sum_v \Phi_{vm}\, \psi(z_v, b_v)$. Typed attention then operates between region anchors. For $r \in \mathcal{R}^{\text{spa}}$ and neighbors $N_r(m) = \{m' : (M_r)_{mm'} > 0\}$,

$$\alpha^r_{mm'} = \frac{\exp\big((W^r_Q s_m)(W^r_K s_{m'})^\top/\sqrt{d'} + b_r\big)}{\sum_{m'' \in N_r(m)} \exp\big((W^r_Q s_m)(W^r_K s_{m''})^\top/\sqrt{d'} + b_r\big)}, \tag{5}$$

$$\tilde{s}_m = \sum_{r \in \mathcal{R}^{\text{spa}}} \sum_{m' \in N_r(m)} \alpha^r_{mm'}\, W_r s_{m'}. \tag{6}$$

Structural relations stay on structural nodes. With $\psi_u = \psi(z_u, b_u, a_u)$ and neighbors $N_r(u)$ under $r \in \mathcal{R}^{\text{str}}$,

$$\tilde{t}_u = \sum_{r \in \mathcal{R}^{\text{str}}} \sum_{u' \in N_r(u)} \alpha^r_{uu'}\, W_r \psi_{u'}, \tag{7}$$

where $\alpha^r_{uu'}$ is the same softmax attention with a relation bias b_r.

Cross-view attention injects the structural state into each anchor. The query is the spatial anchor state $\tilde{s}_m$. The key concatenates the structural state with the box offset $b_u - B_m$ and the remaining attributes a_u, so the score can tell a text match whose rendered box has moved. The coupling $\tilde{Q}$ enters as a log prior and as the support of the sum: a structural node with $\tilde{Q}_{mu} = 0$ is excluded.

$$\ell_{mu} = \frac{(W_Q \tilde{s}_m)^\top W_K [\tilde{t}_u;\, b_u - B_m;\, a_u]}{\sqrt{d}} + \log \tilde{Q}_{mu}, \tag{8}$$

$$c_m = \sum_{u:\, \tilde{Q}_{mu} > 0} \operatorname{softmax}(\ell_{m\cdot})_u\, W_V \tilde{t}_u. \tag{9}$$

The spatial state $\tilde{s}_m$ is kept, and c_m is the structural message received by the region. The two vectors are not concatenated.

The write onto visual tokens is a second cross-attention. Token i with hidden state h_i and box b_i forms the query $q_i = W_Q h_i$. Let $\omega_{im} = \text{IoU}(b_i, B_m)$ and

$$P_{im} = \frac{\exp(\beta_P\, \omega_{im})}{\sum_{m'} \exp(\beta_P\, \omega_{im'})}, \tag{10}$$

with a fixed inverse temperature β_P. The two channels attend separately, and the value of each attention is the corresponding anchor state,

$$\begin{aligned} \delta^V_i &= \sum_m \operatorname{softmax}\big(q_i^\top W_K \tilde{s}_m/\sqrt{d'} + \log P_{im}\big)_m\, W_V \tilde{s}_m, \\ \delta^T_i &= \sum_m \operatorname{softmax}\big(q_i^\top W_K c_m/\sqrt{d'} + \log P_{im}\big)_m\, W_V c_m. \end{aligned} \tag{11}$$

The third message δ^L_i is obtained by encoding the serialized nodes and edges of G. None of the three values is a copy of h_i.

These messages are written at a fixed set of insertion layers. A full residual would add each message along every direction of the receiving state. Because the grounding head is the linear map $\ell_g(H) = HW_g$, the part of a visual message that lies in the row space of W_g moves the predicted point and violates Equation (1). Orthogonal injection removes that part before the gate is applied. The removed directions are the leading subspace of the frozen visual states on plain screenshots. Text tokens are not read by the grounding head, so their messages are left unchanged.

The subspace is estimated once and then held fixed. The frozen backbone is run on plain screenshots without graph input, and the visual hidden states $H^{(0)}_l$ are collected at each insertion layer l. Their truncated singular value decomposition yields the grounding subspace basis

$$H^{(0)}_l = U_l \Sigma_l V_l^\top, \qquad U_{g,l} = \operatorname{top}_r(U_l) \in \mathbb{R}^{d \times r}, \tag{12}$$

where $U_{g,l}$ stacks the r leading left singular vectors with $r \ll d$. The orthogonal projection onto the complement at layer l is

$$P_{\perp,l} = I_d - U_{g,l} U_{g,l}^\top. \tag{13}$$

For each channel $c \in \{V, T, L\}$ the projected message is

$$\tilde{\delta}^c_i = \mathbb{1}_{\text{vis}}(i)\, P_{\perp,l}\, \delta^c_i + \big(1 - \mathbb{1}_{\text{vis}}(i)\big)\, \delta^c_i, \tag{14}$$

where $\mathbb{1}_{\text{vis}}(i)$ indicates that token i is visual. When $\text{Col}(W_g^\top) \subseteq \text{span}(U_{g,l})$, the projection annihilates the grounding row space, $W_g^\top P_{\perp,l} = \mathbf{0}$, and $\ell_g(h + P_{\perp,l}\delta) = \ell_g(h)$ for every message δ. A convex combination of the projected channels inherits the same invariance, which is why the gate of Section 3.4 mixes $\tilde{\delta}^c_i$ rather than δ^c_i. A soft variant $P_\theta = I_d - U_{g,l}\, \text{diag}(\sigma(s))\, U_{g,l}^\top$ relaxes the hard mask when the grounding boundary is uncertain, where $s \in \mathbb{R}^r$ is a learned logit vector and $\sigma(s) \in [0,1]^r$ is the elementwise sigmoid. The readout then changes by at most $\|W_g^\top P_\theta \delta\|_2 \le (1 - \sigma_{\min})\|W_g\|_2\, \|\delta\|_2$, which recovers the hard projection when $\sigma(s) = \mathbf{1}$. When the estimated basis covers the head weights with an angular error α_g, the residual leakage is bounded by $\|W_g^\top P_{\perp,l}\delta\|_2 \le \|W_g\|_2 \sin\alpha_g\, \|\delta\|_2$. The projection is also the least invasive feasible write, and the proof follows from the Lagrange multiplier conditions in Appendix A.

The write is repeated at several insertion layers with a separate basis for each layer. The point is invariant at every layer, and composing the writes preserves this invariance. If the angular error at layer l is $\alpha_{g,l}$, the accumulated leakage across L insertion layers is bounded by

$$\Big\| \ell_g\big(H^{(L)}\big) - \ell_g(H) \Big\|_2 \le \sum_{l=1}^{L} c_l \sin\alpha_{g,l}\, \|\delta_l\|_2, \tag{15}$$

where c_l is the gain of the later layers along the point direction, with the derivation in Appendix A. A fixed residual scale attenuates the structure signal together with the directions that move the point. The projection removes only those directions.

3.4 Reliable Routing

The projected messages of Section 3.3 still differ in what they answer. Spatial relations are read from the rendered page, and sibling order is read from the structure. A fixed gate cannot capture this difference. We first measure channel reliability under a controlled flip, and then train a gate that matches it. For a relation r and a channel $c \in \{V, T, L\}$, the instrument constructs a paired page $\check{x}_c^r$ in which only the evidence of r in channel c is flipped, while the other channels stay unchanged. The paired answer $\check{y}_c^r$ is the answer that follows the flipped evidence, and we write this operation as $\mathrm{do}(\mathrm{flip}_c^r)$. Let $p(\mathrm{flip}_c^r)$ denote the intervention distribution over paired pages. A channel is reliable for r when its answer follows the flip, and the reliability is the response probability under this intervention,

$$\rho_c^r = \mathbb{E}_{\check{x}_c^r \sim p(\mathrm{flip}_c^r)}\Big[\mathbf{1}\big[\hat{y}(\check{x}_c^r) = \check{y}_c^r\big]\Big], \tag{16}$$

which is a causal quantity rather than a correlation between a channel and the answer.

The gate combines the channels for each visual token and relation type through a temperature scaled softmax over the channel logits,

$$g_{ic}^r = \frac{\exp\big(w^\top \varphi_{ic}^r / \tau_g\big)}{\sum_{c' \in \{V,T,L\}} \exp\big(w^\top \varphi_{ic'}^r / \tau_g\big)}, \qquad \Delta_i = \sum_{c \in \{V,T,L\}} g_{ic}^r \, \tilde{\delta}_i^c, \tag{17}$$

where $\varphi_{ic}^r = [h_i, \tilde{\delta}_i^V, \tilde{\delta}_i^T, \tilde{\delta}_i^L, \rho_c^r, e_i]$ collects the hidden state, the three projected messages, the reliability, and the token type indicator e_i into a feature vector, w is a learnable scoring vector, and τ_g is the gate temperature. The gate is a distribution over channels, and the three weights sum to one. The hidden state is then updated by $h_i' = h_i + \lambda_i \Delta_i$, with a per-token scale λ_i. Each visual $\tilde{\delta}_i^c$ lies in the complement of the grounding subspace, and so does Δ_i. The gate is trained with a proper score so that its optimum is the reliability itself. With the Brier score, the squared error between the gate and the flip indicator is minimized by the conditional mean of that indicator, which is exactly the reliability ρ_c^r. When the paired pages are sampled independently and stratified by relation type, the optimal gate satisfies $g_c^* = \rho_c^r$, and with n_p paired pages the empirical estimate obeys $|g_c^* - \hat{\rho}_c^r|_2 \le O(n_p^{-1/2})$, with proof in Appendix A.

The paired pages are generated from WebForge by a small set of edit operators that each act on one channel, including visual order flips, structural order flips, anchor permutations, and edge rewiring. On real pages, a conflict slice enters the set when the rendered order and the structural order disagree under Kendall's τ. The instrument exists only during training and is removed at inference, with details in Appendix E.

3.5 Training Objective

The model is trained with four terms. The supervision term learns the answer, the grounding term fixes the point, the consistency term trains the gate, and the alignment term coordinates the two views. The full objective is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{SFT}}(\theta) + \alpha\,\mathcal{L}_{\text{ground}}(\theta) + \eta\,\mathcal{L}_{\text{cons}}(\theta) + \mu\,\mathcal{L}_{\text{align}}(\theta), \quad (18)$$

where α, η, μ are positive weights. The supervision term is the expected cross entropy of the answer tokens on WebForge pages,

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(G,q,y)\sim\mathcal{D}}\left[\sum_{o}\log p_\theta(y_o \mid y_{<o}, M, G, q)\right]. \quad (19)$$

The grounding term keeps the point distribution fixed on plain pages. Writing $p_g(h) = \text{softmax}(\ell_g(h))$ for the grounding distribution over a coordinate grid, it is the KL divergence between the grounding distribution after and before the write,

$$\mathcal{L}_{\text{ground}} = \mathbb{E}_{G\sim\mathcal{D}_0}\left[\text{KL}\big(p_g(h') \,\big\|\, p_g(h)\big)\right], \quad (20)$$

where $\mathcal{D}_0$ is the distribution of plain pages. The consistency term trains the gate from the paired pages. Let c^+ be the channel that follows the flip and c^- a channel that does not. The hinge rank loss

$$\mathcal{L}_{\text{cons}} = \mathbb{E}_{(r,c^+,c^-)\sim\mathcal{D}_{\text{pair}}}\left[\max\big(0,\ \gamma - g^r_{c^+} + g^r_{c^-}\big)\right] \quad (21)$$

raises the weight of the reliable channel above the others by a margin γ, where $\mathcal{D}_{\text{pair}}$ is the distribution over relation, reliable channel, and unreliable channel triples. The alignment term is the optional InfoNCE contrast between summaries s^{spa} and s^{str} of the two views with temperature κ, and Appendix D derives the mutual information lower bound. The term is switched off when the two views are not aligned by contrastive supervision.

Every layer that writes a message is initialized to zero. The residual is zero at the first step, so training starts from the frozen backbone and the point is unchanged. The cost of the forward pass is given in Appendix B.

4 Experiments

4.1 Experimental Setup

4.1.1 Benchmarks. The main evaluation uses three public benchmarks, and we do not aggregate scores across them because the tasks, units, and scoring functions differ. WebSRC [2] is a web-based structural reading comprehension benchmark, and we use the official DEV subset with two questions per page. We report the two official metrics, exact match (EM) and token-level F1, and a project-internal path overlap score (POS) that asks whether the answer lands on the right node in the page tree. VisualWebBench [15] contains 1536 test instances in seven tasks at three levels, including caption, WebQA, heading OCR, element OCR, element grounding (EG), action prediction, and action grounding (AG). ScreenSpot-v2 [27] evaluates point grounding, where a predicted point is correct when it falls inside the annotated element box. The main table uses the Web subset only, split by the element class into Text and Icon. The GUIAct web subset from GUICourse [1] overlaps with ScreenSpot-v2 Web in both unit and scoring, so it is reported only as a supplementary table in Appendix **??**. Synthetic WebForge pages are used solely as the controlled intervention instrument (Section 3.4), never as main results.

4.1.2 Baselines. To verify the effectiveness of the proposed GPGR, we compare it with the following baseline methods, including the frozen backbone [20], LoRA fine-tuning [10], TLG [6], LAPDoc [12], DOM and A11y prompting [35], Set-of-Mark [30], ShowUI [13], DocLLM [25], DRS [16], and FocusUI [17]. The detail of the baseline models is in Appendix C.

4.1.3 Implementation Details. GPGR freezes the backbone Qwen3-VL-8B-Instruct and trains only the graph plugin, which contains about 5.4M parameters. The plugin uses at most $K^{\max} = 48$ anchors, a 320-dimensional bottleneck, 4 heads, and 2 propagation rounds, and is inserted at language-model layers 3, 11, 19, and 27. A rank-16 grounding subspace is fit once by SVD on 32 pages and then held fixed. Visual residuals stay at zero, so messages are written only onto text tokens, and pages without a document tree are left unchanged. Training is 10 epochs of answer supervision on 85,694 question-answer pairs from 10,000 WebForge pages, with AdamW at learning rate 10^{-3}, micro-batch 24, sequence length 3072, bfloat16, and gradient checkpointing. 512 counterfactual probes add a Brier loss on the router and are not mixed into the answers.

4.2 Performance Comparison

Table 1 reports GPGR against eleven baselines, including the frozen backbone, on WebSRC, VisualWebBench, and the Web split of ScreenSpot-v2. GPGR ranks first on every metric. Where the frozen model is already accurate, as on token F1 and element grounding, the margin is small. The gain is clearer when the answer depends on which region supports it. DOM is the closest reader. Serializing the page tree supplies the relations behind exact match, token F1, POS, and WebQA, and DOM remains second on all four. The prompt still leaves the backbone to align that text with the screenshot, so the same evidence yields a smaller gain and slightly lowers element grounding. On icons, where the target carries little readable text, DOM is also the nearest baseline, and GPGR still leads. DocLLM is the contrasting failure. Trained on the same answers without the tree, it moves exact match only slightly, then loses text grounding, icons, and action grounding, because layout-conditioned attention rewrites spatial features that the frozen readout already used. FocusUI is the localization-side counterpart. It stays close to GPGR on action grounding and text points, yet trails the frozen model on WebSRC, where the discarded context still contains the answer, and captioning drops for the same reason. Structural messages avoid both failures. They improve understanding and choice together, and because they are written onto text tokens outside the grounding subspace, the added evidence does not displace the point readout of the frozen backbone.

4.3 Ablation Study

4.4 Grounding Preservation

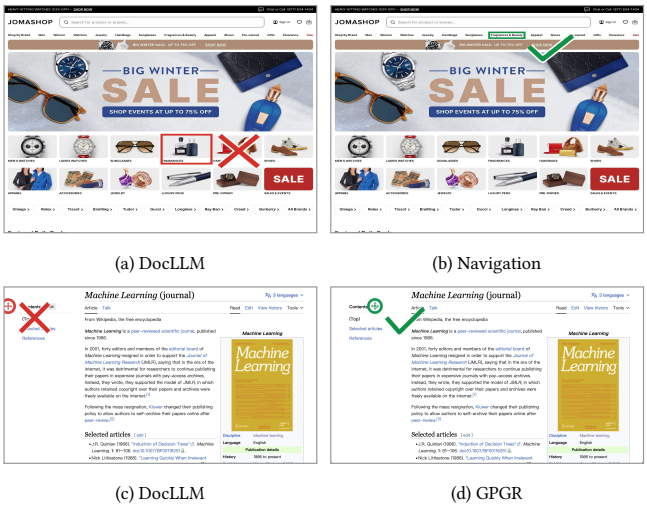
Table 1: Performance comparison of baselines and GPGR. Understanding and Grounding are measured by the corresponding metrics. Best and second-best values are in bold and underlined.

	Metric	Frozen	LoRA	DOM	A11y	TLG	SoM	DocLLM	LAPDoc	ShowUI	DRS	FocusUI	GPGR
WebSRC	EM	86.7	87.3	88.6	85.7	87.2	64.5	87.4	87.0	76.6	67.4	79.8	**89.1**
	F1	90.4	90.8	90.9	89.5	89.8	69.2	90.8	90.2	83.3	71.9	84.2	**91.2**
	POS	77.5	79.2	79.6	78.3	79.4	71.1	77.4	78.3	75.6	68.1	71.5	**81.7**
VisualWebBench	Caption	29.0	30.2	29.9	30.8	28.8	27.3	29.7	31.3	28.9	24.8	25.3	**32.6**
	WebQA	88.0	88.1	89.4	88.7	87.5	84.3	87.6	85.9	84.1	83.7	85.2	**90.4**
	Ele-G	93.0	92.0	91.8	91.5	93.2	92.3	90.6	92.7	93.5	93.9	93.7	**94.9**
	Act-G	87.4	87.4	89.3	88.3	87.4	86.4	83.5	89.3	86.4	90.3	90.3	**91.3**
ScreenSpot-v2	W-Text	92.3	88.4	91.8	91.4	88.0	90.9	88.2	93.5	87.3	94.8	94.7	**95.5**
	W-Icon	85.4	84.9	86.7	86.3	75.9	84.3	72.7	85.6	83.6	85.4	86.1	**87.2**

Table 2: Ablation of GPGR across the three benchmarks.

	WebSRC			VisualWebBench				ScreenSpot-v2	
	EM	F1	POS	WebQA	OCR	EG	AG	Text	Icon
GPGR									
Ablation study on injection									
w/o $P_{\perp}$									
w/o λ									
Ablation study on routing									
w/o g									
w/o ρ									
w/o $\mathcal{L}_{\text{cons}}$									
Ablation study on channel									
w/o V									
w/o T									
w/o L									

4.5 Case Study



(a) DocLLM (b) Navigation

(c) DocLLM (d) GPGR

Figure 2: Case study between DocLLM [25] and GPGR. (a) and (b) come from VisualWebBench (`webqa_176`) with query "Which category is directly to the right of 'Sunglasses' in the navigation menu?". (c) and (d) come from ScreenSpot-v2 (`ssv2_web_1152`) with instruction "hide sidebar".

5 Conclusion and Future Work

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A Theoretical Analysis

This appendix gives the derivations referenced in Section 3. We state each result in the form used in the main text and give the proof in natural prose with the supporting inequalities displayed inline.

A.1 Entropic optimal transport alignment

In this section, we derive the alignment in Eq. (2). Let a and b be strictly positive probability vectors and let every entry of Ψ be finite. The transport polytope is

$$\Pi(a,b) = \left\{ Q \in \mathbb{R}_+^{n^{\mathrm{spa}} \times n^{\mathrm{str}}} \;\middle|\; Q\mathbf{1}_{n^{\mathrm{str}}} = a, \; Q^\top \mathbf{1}_{n^{\mathrm{spa}}} = b \right\}. \tag{22}$$

The independent coupling $a \otimes b$ is defined by $(a \otimes b)_{vu} = a_v b_u$. It is feasible, so the polytope is nonempty, and it is compact and convex. Every feasible matrix has total mass one. Hence the KL term in the main objective is

$$\begin{aligned} \mathrm{KL}(Q \| a \otimes b) &= \sum_{v,u} Q_{vu} \log \frac{Q_{vu}}{a_v b_u} \\ &= \sum_{v,u} Q_{vu} \log Q_{vu} - \sum_v a_v \log a_v - \sum_u b_u \log b_u, \end{aligned} \tag{23}$$

where $0 \log 0 = 0$. Define $\mathcal{H}(Q) = -\sum_{v,u} Q_{vu}(\log Q_{vu} - 1)$. Because the marginals and total mass are fixed, the objective equals

$$\begin{aligned} \langle Q, \Psi \rangle_F + \tau_Q \mathrm{KL}(Q \| a \otimes b) = \langle Q, \Psi \rangle_F - \tau_Q \mathcal{H}(Q) \\ + \tau_Q c(a,b). \end{aligned} \tag{24}$$

Here $c(a,b) = 1 - \sum_v a_v \log a_v - \sum_u b_u \log b_u$ is independent of Q, establishing equivalence to entropy regularization. Continuity on the compact feasible set ensures existence. Strict convexity of $x \log x$ and $\tau_Q > 0$ ensure uniqueness. The minimizer is strictly positive. To see this, interpolate any feasible matrix with a zero entry toward the strictly positive coupling $a \otimes b$. The entropy contribution at the zero entries has a negative directional derivative of unbounded magnitude, whereas the finite cost contributes a bounded derivative. A matrix with a zero entry therefore cannot minimize the objective.

For positive Q, introduce row and column multipliers ξ and ζ. Using the equivalent entropy objective gives the Lagrangian

$$\begin{aligned} \mathscr{L}(Q, \xi, \zeta) = \sum_{v,u} \left[\Psi_{vu} Q_{vu} + \tau_Q Q_{vu} (\log Q_{vu} - 1) \right] \\ + \xi^\top (a - Q\mathbf{1}) + \zeta^\top (b - Q^\top \mathbf{1}). \end{aligned} \tag{25}$$

Stationarity yields $\Psi_{vu} + \tau_Q \log \hat{Q}_{vu} - \xi_v - \zeta_u = 0$. Writing $\mathcal{K}_{vu} = \exp(-\Psi_{vu}/\tau_Q)$, $p_v = \exp(\xi_v/\tau_Q)$, and $t_u = \exp(\zeta_u/\tau_Q)$ gives

$$\hat{Q} = \mathrm{diag}(p) \mathcal{K} \, \mathrm{diag}(t), \qquad p > 0, \quad t > 0. \tag{26}$$

Here p and t are scaling vectors rather than node identifiers. The factors absorb the marginal terms that would appear if the Lagrangian were written directly for the KL objective. Enforcing the marginals leads to the Sinkhorn updates

$$p^{(j+1)} = a \oslash (\mathcal{K} t^{(j)}), \qquad t^{(j+1)} = b \oslash (\mathcal{K}^\top p^{(j+1)}), \tag{27}$$

where $\oslash$ denotes elementwise division and $t^{(0)} > 0$. For a strictly positive kernel, the scaled matrices converge to the unique coupling [4]. Each update satisfies one marginal exactly, while both hold jointly at convergence. Intermediate iterates need not lie in $\Pi(a,b)$. Each dense iteration costs $O(n^{\mathrm{spa}} n^{\mathrm{str}})$ arithmetic operations. A stable implementation performs the updates in the logarithmic domain when the regularization parameter is small.

The resulting coupling is soft and generally dense. Zero entries cannot be assumed without an additional support constraint or an explicit sparsification step.

A.2 Properties of anchor aggregation

The assignment $\Phi \in \mathbb{R}^{n^{\mathrm{spa}} \times K}$ is nonnegative and satisfies $\Phi \mathbf{1}_K = \mathbf{1}_{n^{\mathrm{spa}}}$, including rows assigned by the nearest anchor fallback. Since $\tilde{Q} = \Phi^\top \hat{Q}$, the structural marginal obeys

$$\begin{aligned} \tilde{Q}^\top \mathbf{1}_K &= \hat{Q}^\top \Phi \mathbf{1}_K \\ &= \hat{Q}^\top \mathbf{1}_{n^{\mathrm{spa}}} = b. \end{aligned} \tag{28}$$

Thus aggregation redistributes spatial alignment mass among anchors without changing the total mass associated with any structural node. The anchor marginal is $\tilde{Q}\mathbf{1}_{n^{\mathrm{str}}} = \Phi^\top a$, and the total mass remains one. These identities concern alignment mass and do not establish that aggregation preserves all node attributes or relations.

For each spatial relation r, let $A^r \in \mathbb{R}_+^{n^{\mathrm{spa}} \times n^{\mathrm{spa}}}$ be its adjacency matrix on the spatial nodes. The corresponding matrix between anchors is

$$M_r = \Phi^\top A^r \Phi \in \mathbb{R}_+^{K \times K}. \tag{29}$$

Expanding the matrix product gives

$$(M_r)_{mm'} = \sum_{v,v'} \Phi_{vm} A^r_{vv'} \Phi_{v'm'}. \tag{30}$$

Each original relation contributes to an anchor pair according to the assignments of its endpoints. Under a hard assignment, the entry sums the relation weights between nodes assigned to the two anchors. Under a soft assignment, the contribution is distributed across anchor pairs. In either case, nonnegativity and the row sums of Φ imply

$$\sum_{m,m'} (M_r)_{mm'} = \sum_{v,v'} A^r_{vv'}. \tag{31}$$

The identity includes diagonal entries, which represent relations whose endpoints contribute to the same anchor. Consequently, $N_r(m) = \{m' \mid (M_r)_{mm'} > 0\}$ identifies anchor pairs supported by the original relation and the assignment. This support is used by the typed attention in Section 3.3. An entry summarizes relations between constituent nodes and does not assert that the enclosing anchor boxes themselves satisfy relation r.

A.3 Exact grounding invariance

Let $W_g \in \mathbb{R}^{d \times 2}$ be the linear grounding head and let $U_{g,l} \in \mathbb{R}^{d \times r}$ be the estimated grounding basis at layer l from Equation (12), with complement projection $P_{\perp,l} = I_d - U_{g,l} U_{g,l}^\top$ from Equation (13). If the grounding row space satisfies $\mathrm{Col}(W_g^\top) \subseteq \mathrm{span}(U_{g,l})$, then each column of W_g lies in $\mathrm{span}(U_{g,l})$, so $P_{\perp,l} W_g = \mathbf{0}$ and equivalently $W_g^\top P_{\perp,l} = \mathbf{0}$. For any message δ written on visual tokens, the readout after injection satisfies

$$\ell_g(h + P_{\perp,l}\delta) = (h + P_{\perp,l}\delta) W_g = h W_g + \delta^\top P_{\perp,l}^\top W_g = h W_g = \ell_g(h), \tag{32}$$

which is the exact invariance stated in Section 3.3. For the soft projection $P_\theta = I_d - U_{g,l}\,\mathrm{diag}(\sigma(s))\,U_{g,l}^\top$ with $\sigma(s) \in [0,1]^r$, the residual reads $W_g^\top P_\theta \delta = W_g^\top U_{g,l}\,\mathrm{diag}(\sigma(s))\,U_{g,l}^\top \delta$. Taking norms and using $\|W_g^\top U_{g,l}\|_2 \le \|W_g\|_2$ together with $\|\mathrm{diag}(\sigma(s))\|_2 \le 1 - \sigma_{\min}$ gives

$$\|W_g^\top P_\theta \delta\|_2 \le (1 - \sigma_{\min})\,\|W_g\|_2\,\|\delta\|_2, \tag{33}$$

which is the soft projection bound, and it recovers the hard projection when $\sigma(s) = \mathbf{1}$.

A.4 Minimal safe perturbation

Write $\mathcal{G}_l = \mathrm{span}(U_{g,l})$ and consider the constrained least squares problem

$$\min_{z \in \mathcal{G}_l^\perp} \|z - \delta\|_2^2 \quad \text{s.t.} \quad U_{g,l}^\top z = \mathbf{0}. \tag{34}$$

The Lagrangian is $\mathscr{L}(z,\nu) = \frac{1}{2}\|z-\delta\|_2^2 + \nu^\top U_{g,l}^\top z$, where $\nu \in \mathbb{R}^r$ collects the multipliers. The stationarity condition gives $z - \delta + U_{g,l}\nu = \mathbf{0}$, and the constraint gives $U_{g,l}^\top z = \mathbf{0}$. Substituting $z = \delta - U_{g,l}\nu$ into the constraint yields $U_{g,l}^\top \delta - U_{g,l}^\top U_{g,l}\nu = \mathbf{0}$, so $\nu = (U_{g,l}^\top U_{g,l})^{-1} U_{g,l}^\top \delta$ when the columns of $U_{g,l}$ are orthonormal, which gives the minimizer

$$z^* = \delta - U_{g,l} U_{g,l}^\top \delta = P_{\perp,l}\,\delta. \tag{35}$$

Every feasible z satisfies $W_g^\top z = \mathbf{0}$ because $\mathrm{Col}(W_g^\top) \subseteq \mathcal{G}_l$. When $\mathrm{Col}(W_g^\top) = \mathcal{G}_l$, the null space of W_g equals $\mathcal{G}_l^\perp$, so the constrained problem $\arg\min_{z:\, W_g^\top z = \mathbf{0}} \|z - \delta\|_2^2$ has the same feasible set and the same unique minimizer $P_{\perp,l}\delta$.

A.5 Angular leakage and the bound across layers

Let $\alpha_{g,l} \in [0, \pi/2]$ be the principal angle between $\mathrm{Col}(W_g^\top)$ and $\mathrm{span}(U_{g,l})$, so the component of W_g outside the estimated subspace has norm bounded by $\|W_g\|_2 \sin\alpha_{g,l}$. For a message δ_l written at layer l, the leakage through the estimated projection satisfies

$$\|W_g^\top P_{\perp,l}\delta_l\|_2 \le \|W_g\|_2 \sin\alpha_{g,l}\,\|\delta_l\|_2. \tag{36}$$

When the writes are composed across L insertion layers, a message leaking at layer l is amplified by the gain c_l of the later layers along the point direction. Telescoping the per-layer leakage through the gains of the later layers yields the accumulated bound in Equation (15), $\left\|\ell_g(H^{(L)}) - \ell_g(H)\right\|_2 \le \sum_{l=1}^{L} c_l \sin\alpha_{g,l}\,\|\delta_l\|_2$. The soft projection bound is the special case in which the angular error is replaced by the mask slack $1 - \sigma_{\min}$.

A.6 Reliability is the optimal gate

Let $Z_c^r \in \{0,1\}$ be the flip indicator for channel c and relation r, equal to one when the answer follows the flipped evidence. The Brier objective $\mathbb{E}_{(r,c)}\left[(g_c^r - Z_c^r)^2\right]$ is minimized pointwise in g_c^r by the conditional mean

$$g_c^* = \mathbb{E}[Z_c^r \mid r, c] = \rho_c^r, \tag{37}$$

which is the reliability defined in Equation (16). When the paired pages are sampled independently and stratified by relation type, Hoeffding's inequality gives

$$\Pr\left(|\hat{\rho}_c^r - \rho_c^r| > \epsilon\right) \le 2\exp(-2n_p\,\epsilon^2), \tag{38}$$

so $|\hat{\rho}_c^r - \rho_c^r| = O(n_p^{-1/2})$ with high probability. The softmax gate in Equation (17) is Lipschitz in its logits with constant bounded by $1/\tau_g$, so the same rate carries to the learned gate weights, giving $|g_c^* - \hat{\rho}_c^r|_2 \le O(n_p^{-1/2})$.

B Complexity

The forward pass never attends over token pairs. Spatial attention on the K anchors costs $O(K^2|\mathcal{R}^{\mathrm{spa}}|)$. Structural attention costs $O(|E^{\mathrm{str}}|)$, linear in the number of structural edges. Cross-view attention costs $O(K\,n^{\mathrm{str}})$ and drops to the support of $\tilde{Q}$ once zero-mass structural nodes are skipped. The write from anchors to visual tokens costs $O(nK)$, and the gate costs $O(d)$. The grounding subspace is estimated once per backbone at $O(ndr)$. Each insertion layer applies the projection at $O(ndr)$ with $r \ll d$. The pass therefore scales as $O(nK + Kn^{\mathrm{str}} + K^2|\mathcal{R}^{\mathrm{spa}}|)$ with the number of visual tokens, whereas full attention over a token graph scales as $O(n^2)$. The Sinkhorn alignment is computed once before training. The intervention instrument is removed at inference and adds no serving cost.

C Baseline Descriptions

We compare GPGR with the following state-of-the-art methods.

- **Frozen** [20]. The frozen backbone with no additional training, evaluated zero-shot.
- **LoRA** [10]. The frozen backbone with LoRA fine-tuning on the same WebForge data and budget, but no structural representation.
- **TLG** [6]. Graph text in the Talk-like-a-Graph encoding, trained under the shared LoRA budget.
- **LAPDoc** [12]. Region boxes verbalized in the LAPDoc format as a prompt, training-free.
- **DOM** [35]. The HTML DOM serialized as a prompt, training-free.
- **A11y** [35]. The accessibility tree serialized as a prompt, training-free.
- **SoM** [30]. Set-of-Mark prompting that annotates the screenshot with bounding boxes, training-free.
- **ShowUI** [13]. The screenshot formulated as a connected UI graph for token selection, training-free.
- **DocLLM** [25]. Layout-conditioned attention adapted under the shared LoRA budget.
- **DRS** [16]. A training-free region search that proposes instruction-relevant regions on the screenshot and prunes the remaining interface elements. The page tree is not read.
- **FocusUI** [17]. Instruction-relevant visual tokens are selected with a score that combines the query and a UI-graph saliency prior, and a position marker keeps the order of the retained patches. The page tree is not read.

D Cross-View Alignment Derivation

[TODO] InfoNCE lower bound $I(s_\theta(G^{\mathrm{spa}}); s_\theta(G^{\mathrm{str}})) \ge \log N - L_{\mathrm{NCE}}$ [23] and the tighter HSIC/kernel-dependence form [7]; used only when $\mathcal{L}_{\mathrm{align}}$ is enabled.

E Intervention Instrument Details

[TODO] WebForge template families and edit rules; paired conflict examples; real-page Kendall τ slicing; annotator and scoring details; deterministic scoring without model judges.